

VOLUME I

Platform, Data Sources and Evidence Storage

A code-grounded explanation of the problem, architecture, capital stack, online sources, acquisition controls, parsing, lineage, and database design.

FOCUS

Where the data comes from, how it is retrieved and parsed, what is stored, and how an auditor traces every ranked input back to immutable evidence.

Implementation snapshot: 02 September 2026

Implemented behavior is separated from planned production extensions on every page.

Contents and reading guide

Read this volume first. It establishes the economic question, the implemented platform boundary, every online source, the ingestion and parsing path, and the complete persistence model used by Volumes II and III.

Status convention

IMPLEMENTED means executable behavior in this repository. PLANNED means a designed production extension that is catalogued or declared but not wired. BOUNDARY means a deliberate non-goal or limitation.

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Reference capital-stack image

The following user-supplied image is the sole origin of the seed spend amounts in taxonomy version 2026.09.01. The platform treats the normalized weights as unvalidated discovery hypotheses, not audited market size.

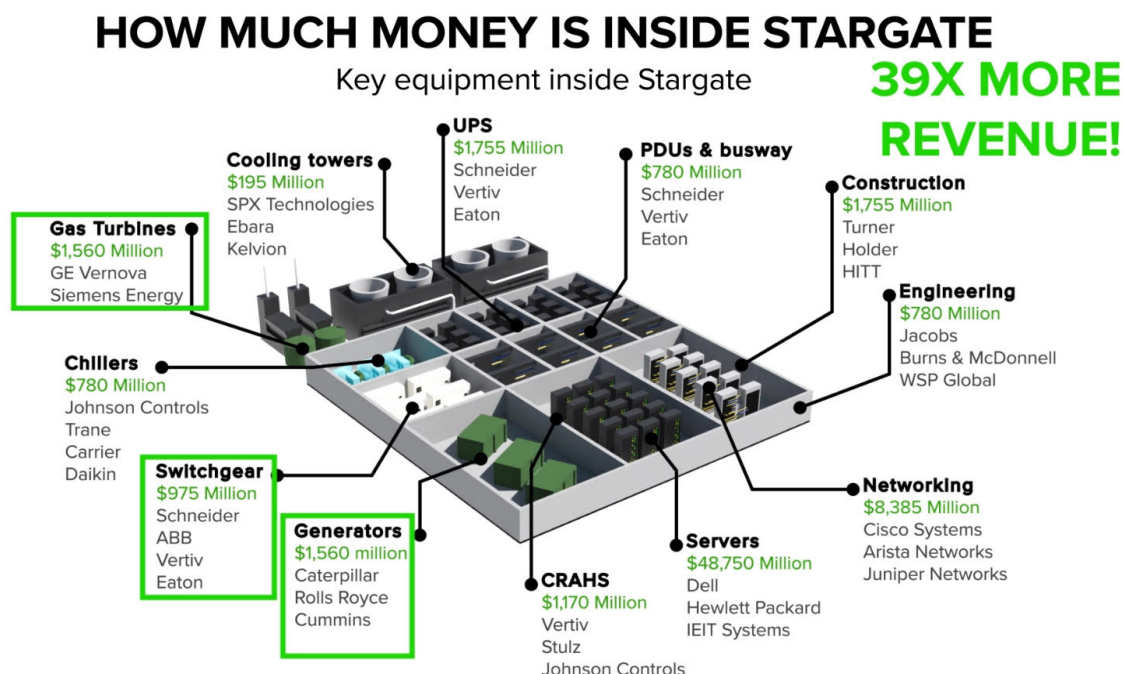


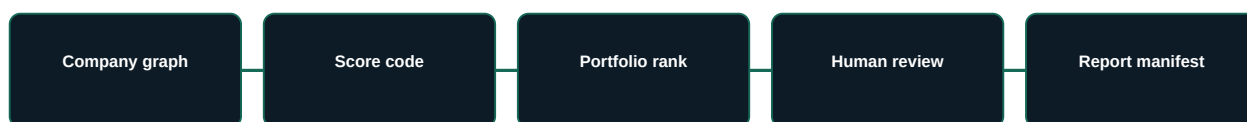
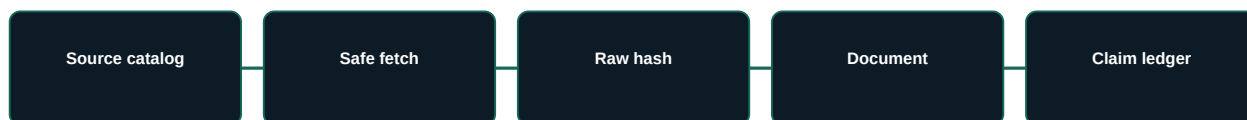
Figure 1. User-provided Stargate equipment and capital allocation reference.

Interpretation rule

These values guide value-chain mapping. They are not multiplied into company TAFGS and must be replaced or corroborated by independently licensed market evidence for production research.

Architecture at a glance

The platform moves from untrusted external content to approved evidence through explicit boundaries. Every stage narrows authority rather than expanding it.



Key invariant

No raw webpage, filing, news result, or model response can write a final score or published report directly.

SECTION 01

The decision problem

IMPLEMENTED

The platform answers a narrow investment-research question: which public companies have the strongest fundamental growth exposure to AI-factory and hyperscale data-center capital expenditure over a three-year horizon?

The problem is not a web-search problem and not a generic stock screener. It is an evidence-integration problem across different capital layers, issuers, reporting systems, time periods, units, and source rights. The system must convert that heterogeneous evidence into comparable company dossiers without allowing a language model to invent a missing input.

The result is a ranked research output, not an expected-return forecast. Valuation, entry price, discount rates, financing timing, and market sentiment are intentionally absent. A high score means strong, defensible and material AI-factory-driven operating growth under the methodology; it does not mean the security is cheap or will outperform.

The design therefore prioritizes identity, point-in-time evidence, lineage, deterministic arithmetic, uncertainty visibility, and human publication control. Those priorities explain nearly every architecture choice described in these volumes.

Implementation details

- Freeze one research cutoff date for each run.
- Map every issuer to one defined capital-stack role and selected public security.
- Require cited evidence for exposure, margin, growth scenarios, moat, and risk.
- Rank only dossiers that clear identity, coverage, confidence, contradiction, and numeric-validation gates.

Why this design

A narrower claim can be tested and governed. Mixing fundamental growth, valuation, macro timing, and trading signals into one autonomous agent would make errors harder to detect and outputs harder to reproduce.

SECTION 02

System objectives and non-goals

BOUNDARY

The objective is repeatable quarterly equity-research automation with traceable evidence, not unconstrained autonomous investing.

The implemented platform maps the AI-factory capital stack, builds a source-derived public-company universe, stores immutable source lineage, converts documents into deterministic facts or review-only proposals, runs a bounded research graph, computes TAFGS in code, and applies human review before publication.

Phase-one non-goals are equally important: no private companies, venture-stage suppliers, short-term trading signals, portfolio construction, valuation timing, order execution, macro-rate forecasts, autonomous external messages, or unreviewed model-authored claims. These boundaries keep the agent tool set read-only and the blast radius small.

In scope	Explicitly out of scope
Equity research automation	Order execution or trading
AI-factory capex attribution	Valuation and entry price
Moat, margin, growth and risk	Macroeconomic prediction
Top-20 fundamental ranking	Guaranteed stock-price return
Quarterly reproducible refresh	Private-company coverage

SECTION 03

Research questions the platform must prove

IMPLEMENTED

Every rankable company dossier must answer five proof questions with dated evidence and explicit calculations.

First, the system must prove that the legal entity and selected public security are correct for the research date. Second, it must show how the issuer monetizes a defined AI-factory layer. Third, it must quantify the current revenue exposure rather than confusing total addressable market with company revenue. Fourth, it must assess differentiation, margin quality, growth drivers, and risks under versioned rubrics. Fifth, it must allow an analyst or auditor to reproduce the numeric result from the stored claims and configuration snapshot.

No single source can answer all five. Regulatory filings are strong for identity and reported financials but weak for a standardized AI-revenue tag. Project records may show demand but not supplier revenue. Technical standards may show influence but not realized economics. The architecture uses a portfolio of sources and keeps discovery, evidence, judgment, and calculation as separate stages.

Implementation details

- Who is the issuer and which security is eligible?
- Where in the capital stack does the issuer earn revenue?
- What share of company revenue is directly exposed today?
- What evidence supports moat, margin, scenarios, and risk?
- Can a reviewer reproduce the score and publication decision?

SECTION 04

Architecture principles

IMPLEMENTED

Eight principles turn a potentially open-ended research agent into a governed evidence system.

Principle	Concrete implication
Point-in-time by construction	Every run has an as_of_date; later filings and restatements are excluded.
Evidence before judgment	Agents consume normalized claims and evidence IDs, not unrestricted browsing context.
Workflow before autonomy	A typed graph fixes role order, state, tools, retry boundaries, and stopping conditions.
Code owns numbers	Exposure conversion, margins, risk discount, TAFGS, ranking, and release gates are deterministic.
Confidence is separate	Source confidence controls evidence fitness; it never becomes a business-quality score.
Retrieved text is untrusted	Document instructions are scanned and quoted as data, never followed.
Humans authorize publication	Model proposals and ranked dossiers require explicit review.
Reproducibility over mutation	Runs, hashes, versions, and outputs remain traceable instead of being silently overwritten.

Why this design
These principles are stronger than a prompt. They are encoded in schemas, database constraints, state transitions, security checks, deterministic functions, and publication gates.

SECTION 05

End-to-end component architecture

IMPLEMENTED

The platform is arranged as six cooperating planes: sources, acquisition, knowledge, agent workflow, deterministic control, and human release.

The source plane contains regulatory, issuer, project, technical, and discovery providers. The acquisition plane validates a versioned source definition, performs bounded HTTPS retrieval, archives the original response, and normalizes content. The knowledge plane persists company identity, documents, atomic claims, and the capital-stack taxonomy.

The company-research graph reads a frozen evidence bundle and passes a typed assessment through specialized roles. The control plane performs range checks, scoring, scenario ranking, evaluation, review, reporting, and publication. Telemetry crosses all planes through JSON logs, process metrics, and persistent audit events.

Concrete contract

```
SOURCES -> SAFE FETCH -> RAW SNAPSHOT -> NORMALIZED DOCUMENT
-> CLAIM LEDGER -> COMPANY GRAPH -> DETERMINISTIC SCORE
-> PORTFOLIO RANK -> ANALYST REVIEW -> REPORT + MANIFEST
```

Why this design

Each arrow is a contract boundary. Raw content cannot jump directly into ranking, and a model response cannot bypass the proposal ledger or deterministic scorer.

SECTION 06

Reference runtime topology

IMPLEMENTED

The repository implements a real, reproducible single-node topology suitable for an assignment, developer workstation, or controlled pilot.

One Python process composes the FastAPI service, connector services, model gateway, LangGraph or local graph, deterministic scoring services, and report generator. SQLite runs in WAL mode. Raw responses and reports live in local artifact directories. A bounded ThreadPoolExecutor fans out independent company workflows; the steps within one company remain sequential.

The topology minimizes setup cost and makes the complete workflow inspectable. It is intentionally not a multi-writer distributed system. The design documents the migration seams - repository interfaces, raw snapshot abstraction, model gateway, and outer workflow boundary - that should be implemented before horizontal scale.

Concern	Reference implementation
API and workbench	FastAPI + dependency-free HTML
Research graph	LangGraph when installed; local sequential fallback
Portfolio concurrency	ThreadPoolExecutor, default four workers
System of record	SQLite with WAL and foreign keys
Raw evidence	Content-addressed local filesystem
Reports	Markdown plus SHA-256 manifest

SECTION 07

Target production topology

PLANNED

The intended institutional topology separates request handling, durable workflows, transactional data, immutable objects, and observability.

API replicas authenticate users and enqueue work. Temporal or an equivalent durable orchestrator owns long-running source and quarterly jobs. Worker pools execute source adapters and company graphs with provider-specific rate limits. PostgreSQL becomes the multi-writer system of record; S3 or MinIO stores immutable raw bytes, normalized pages, and report artifacts.

Redis may coordinate distributed rate limits, locks, and short-lived caches, but it never becomes authoritative evidence. OpenTelemetry exports traces and metrics to an organization collector. SSO, RBAC, a secret manager, approved egress, signed images, and a private model gateway complete the production trust boundary.

Implementation details

- Migrate repositories and schema before adding multiple writers.
- Keep the company graph bounded inside a durable outer workflow.
- Use object versioning and retention for raw evidence and manifests.
- Do not place n8n inside the scoring or research-decision path.
- Gate model and licensed-data processing through organizational approvals.

SECTION 08

Repository and module map

IMPLEMENTED

Implementation ownership is explicit: configuration owns policy, modules own mechanisms, and persisted manifests bind versions together.

Path	Responsibility
config/	Taxonomy, source, extraction, scoring, and prompt policies
src/aifactory/sources/	Catalog, HTTPS transport, raw storage, and live adapters
src/aifactory/extraction.py	Chunk retrieval, model proposals, deterministic validation, human conversion
src/aifactory/agents/roles.py	Nine bounded research roles
src/aifactory/workflow.py	LangGraph/local typed company graph
src/aifactory/scoring.py	Exposure-aware TAFGS and scenario ranking
src/aifactory/database.py	SQLite schema and repositories
src/aifactory/service.py	Composition root, portfolio fan-out, review, publication
src/aifactory/api.py / cli.py	Analyst and operator interfaces
docs/ and tests/	Operating specification and executable controls

SECTION 09

Capital-stack model

BOUNDARY

The taxonomy converts a data-center build into mutually understandable research layers and candidate-discovery hypotheses.

The seed model comes from the user-supplied Stargate reference image. The displayed amounts total USD 68.445 billion. Those amounts are normalized into five layer weights and persisted under taxonomy version 2026.09.01. They guide market mapping and candidate discovery; the database marks them unvalidated assumptions until supported by independent market evidence.

The weights are not multiplied directly into TAFGS. The company growth forecast already reflects the addressable end market, product cycle, share, backlog, and execution assumptions. Multiplying the same capital distribution again would double-count opportunity size and mechanically favor compute companies.

Layer	Reference spend	Seed share
Compute and servers	\$48.750B	71.225%
Networking	\$8.385B	12.251%
Power infrastructure	\$6.630B	9.686%
Cooling systems	\$2.145B	3.134%
Engineering and construction	\$2.535B	3.703%

SECTION 10

Capital layer: Compute and servers

BOUNDARY

Reference share 71.225%; configured subsegments: accelerators, cpu, hbm, ai_servers, storage.

Map chip, memory, system, and storage revenue without counting a complete server and each component as separate end-market dollars. Evidence must distinguish shipped products, backlog, capacity, and vendor-reported addressable market.

Architectural software lock-in, accelerator supply, packaging, memory bandwidth, platform design wins, and hyperscaler concentration are typical moat and risk questions.

Candidate selection is global and source-derived. A company is not eligible merely because its brand appears in a diagram; it needs a public security, a defined role, and accepted evidence for the scoring dimensions.

Taxonomy field	Configured value
Segment ID	compute_servers
Reference spend	USD 48.750B
Reference weight	0.71225
Subsegments	Accelerators, CPUs, high-bandwidth memory, AI servers, and storage

SECTION 11

Capital layer: Networking

BOUNDARY

Reference share 12.251%; configured subsegments: switches, nic_dpu, infiniband, ethernet, optics, cabling.

Separate scale-up fabric from scale-out networking and front-end connectivity. Normalize port speeds, generation timing, attach rates, optical reach, and whether revenue is product, silicon, module, or system level.

Reference architectures, protocol leadership, interoperability, qualification cycles, optical capacity, and transition risk from 400G to 800G and beyond are central evidence themes.

Candidate selection is global and source-derived. A company is not eligible merely because its brand appears in a diagram; it needs a public security, a defined role, and accepted evidence for the scoring dimensions.

Taxonomy field	Configured value
Segment ID	networking
Reference spend	USD 8.385B
Reference weight	0.12251
Subsegments	Switches, NIC/DPU, InfiniBand, Ethernet, optics, and cabling

SECTION 12

Capital layer: Power infrastructure

BOUNDARY

Reference share 9.686%; configured subsegments: generation, transformers, switchgear, ups, pdu_busway, generators, energy_storage.

Trace monetization from utility/interconnection need to an issuer's actual order, equipment shipment, service agreement, or backlog. Do not convert regional load forecasts directly into supplier revenue.

Lead times, installed-service networks, qualification, grid constraints, gas-turbine supply, transformer capacity, and customer/project concentration affect both scarcity moat and execution risk.

Candidate selection is global and source-derived. A company is not eligible merely because its brand appears in a diagram; it needs a public security, a defined role, and accepted evidence for the scoring dimensions.

Taxonomy field	Configured value
Segment ID	power
Reference spend	USD 6.630B
Reference weight	0.09686
Subsegments	Generation, transformers, switchgear, UPS, PDU/busway, generators, and storage

SECTION 13

Capital layer: Cooling systems

BOUNDARY

Reference share 3.134%; configured subsegments: liquid_cooling, cdu, chillers, cooling_towers, crah, heat_rejection.

Classify facility cooling and direct-to-chip liquid systems separately. Record rack-density assumptions, retrofit versus greenfield exposure, attachment rates, refrigerant/environmental constraints, and supplier content per MW only when supported.

Thermal performance, validated reference designs, service coverage, manufacturing scale, and rapid architecture change determine whether current scarcity becomes durable differentiation.

Candidate selection is global and source-derived. A company is not eligible merely because its brand appears in a diagram; it needs a public security, a defined role, and accepted evidence for the scoring dimensions.

Taxonomy field	Configured value
Segment ID	cooling
Reference spend	USD 2.145B
Reference weight	0.03134
Subsegments	Liquid cooling, CDUs, chillers, towers, CRAHs, and heat rejection

SECTION 14

Capital layer: Engineering and construction

BOUNDARY

Reference share 3.703%; configured subsegments: design, general_contracting, commissioning, modular_construction.

Use awarded backlog, scope, project phase, expected construction period, geographic mix, and fee/revenue recognition. Avoid treating an announced campus value as the contractor's revenue.

Local labor, permitting, power readiness, fixed-price exposure, schedule penalties, customer concentration, and repeat relationships drive growth quality and risk.

Candidate selection is global and source-derived. A company is not eligible merely because its brand appears in a diagram; it needs a public security, a defined role, and accepted evidence for the scoring dimensions.

Taxonomy field	Configured value
Segment ID	construction
Reference spend	USD 2.535B
Reference weight	0.03703
Subsegments	Design, general contracting, commissioning, and modular construction

SECTION 15

Evidence portfolio strategy

IMPLEMENTED

Identity, reported financials, direct exposure, project demand, differentiation, and risk require different source families.

The platform uses regulatory and government sources first for identity and reported numbers, issuer-primary sources for product and segment detail, project/utility records for independent demand evidence, standards and technical documents for defensibility, and licensed specialist sources where public coverage is insufficient.

Discovery sources identify what to investigate. They do not become material evidence merely because they are timely or easy to query. Each source definition declares its evidence tier, licence note, rate limit, timeout, response cap, authentication reference, and implementation status.

Research question	Preferred evidence
Issuer/security identity	Regulator universe + LEI/security master
Reported revenue/margin	Official XBRL and audited filing
Direct AI exposure	Segment notes, IR, backlog, product disclosures, licensed transcripts
Demand pipeline	Hyperscaler filings, permits, utilities, procurement, project awards
Moat	Technical docs, standards, certifications, design wins, competitor evidence
Risk	Risk factors, concentration notes, export rules, supply agreements, cancellations

SECTION 16

Source tiers and authority

IMPLEMENTED

Tiering determines what a source may prove and how confidence is weighted; it does not turn weak evidence into a business-quality score.

Tier 1 includes regulators, company-primary records, governments, and official project records. Tier 2 covers licensed transcripts, standards bodies, technical documents, and utility records. Tier 3 covers reputable news, industry publications, and analyst discovery.

Evidence confidence is calculated as the mean of claim confidence multiplied by the tier weight. Reported financials require Tier 1. Tier 3 cannot be the sole support for a material numeric claim. Contradictory evidence is retained and blocks rankability until adjudicated.

Tier	Weight	Permitted role
1	1.00	Authoritative identity, reported financials, official facts
2	0.80	Specialist and near-primary evidence
3	0.45	Discovery and corroboration only

Concrete contract

```
weighted_claim_confidence = claim.confidence * source_tier_weight
evidence_confidence = mean(weighted_claim_confidence)
```

SECTION 17

Point-in-time evidence

IMPLEMENTED

A research cutoff is an admissibility rule, not a display label.

The service rejects a future `as_of_date`. Database claim queries include only as-of-eligible claims published on or before the cutoff. SEC filing selection excludes filings after the cutoff, and annual facts are ordered by period end, filing date, and accession. A later restatement must create a new run if it changes the historical view.

Point-in-time identity matters too. Tickers, names, listings, parents, and currency inputs can change. The reference schema captures the selected identity and configuration snapshot for a run; a production security master should add effective dating and corporate-action history before global backtesting.

Implementation details

- Never use database insertion time as the economic date.
- Keep original and corrected document hashes.
- Version currency and classification inputs.
- Recompute through a new immutable run rather than rewriting the old rank.

SECTION 18

Entity and security resolution

IMPLEMENTED

The legal entity, public parent, and tradable security are related but not interchangeable.

The implemented U.S. path uses CIK as the stable issuer key, persists all tickers and exchanges returned by the SEC, creates a selected security record, and optionally resolves a GLEIF LEI. Eligibility requires a public exchange and security identifier. Fuzzy matches may propose candidates but cannot silently merge issuers.

Production coverage must add effective-dated mappings for ADR versus primary listing, dual listings, share classes, mergers, spin-offs, ticker changes, subsidiaries, and public-parent relationships. Ambiguous matches require analyst review and must preserve every candidate and score.

Concrete contract

```
company_id: sec-{zero_padded_CIK}  
security_id: CIK{zero_padded_CIK}  
identifier rows: (company_id, scheme, value, source_id, confidence)
```

SECTION 19

Data rights and legal controls

BOUNDARY

Technical access is not permission to store, process with a model, display, or redistribute data.

Every source catalog entry includes licence notes and an authentication reference. Public sources still have fair-access, attribution, retention, and acceptable-use requirements. Commercial data requires contract language covering machine processing, storage, derived values, user display, reports, and model-provider transfer.

The source-onboarding gate therefore includes a data-owner and legal review before `enabled=true`. The system does not bypass paywalls, CAPTCHAs, robots restrictions, authentication controls, or contractual prohibitions. Free model endpoints are restricted to public evidence and synthetic records unless the organization approves their data policy.

SECTION 20

Data source: SEC EDGAR

IMPLEMENTED

Build the U.S. issuer universe, persist CIK/ticker identity, archive submissions and filings, and create authoritative revenue and operating-margin claims from XBRL.

Catalog ID `sec_edgar` selects the code-owned connector `sec_edgar`. The catalog cannot dynamically import arbitrary code. Base URL `https://data.sec.gov` is HTTPS and the adapter is limited to 8.0 requests per second, a 15-second timeout, and 52,428,800 response bytes.

Authentication handling: No credential. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Public regulatory records; comply with SEC fair-access and declared User-Agent requirements.

Universe payload schema is checked for the exact CIK, name, ticker, and exchange field order before records are accepted. Company sync archives submissions JSON, Company Facts JSON, and selected filing HTML. Filing HTML is normalized after script/style/template removal. Annual revenue candidates are limited to 10-K, 20-F, or 40-F facts filed on or before the cutoff; operating income must match the selected revenue period.

Property	Value
Category	regulatory_filings
Evidence tier	1
Status	implemented
Enabled	True
Rate limit	8.0 requests/second
Timeout / response cap	15 sec / 52,428,800 bytes

Why this design

This adapter is live because it has a bounded implementation and governance contract.

Source and reference links

[SEC EDGAR APIs](https://www.sec.gov/search-filings/edgar-application-programming-interfaces)

<https://www.sec.gov/search-filings/edgar-application-programming-interfaces>

[SEC fair-access resources](https://www.sec.gov/about/developer-resources)

<https://www.sec.gov/about/developer-resources>

SECTION 21

Data source: Global LEI Index

IMPLEMENTED

Resolve a selected company legal name to an LEI and persist a scored identity mapping; it is not used to infer financial or investment facts.

Catalog ID gleif selects the code-owned connector gleif. The catalog cannot dynamically import arbitrary code. Base URL <https://api.gleif.org> is HTTPS and the adapter is limited to 2.0 requests per second, a 30-second timeout, and 10,485,760 response bytes.

Authentication handling: No credential. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Open LEI reference data; retain GLEIF attribution and source timestamp.

Exact legal-name filtering runs first; fuzzy completion is a fallback used only to locate a canonical record. Names are case-folded, punctuation-normalized, and stripped of configured legal suffixes before SequenceMatcher similarity. The LEI is persisted only at or above the configured 0.82 similarity threshold, together with match confidence and the matched legal name.

Property	Value
Category	entity_identity
Evidence tier	1
Status	implemented
Enabled	True
Rate limit	2.0 requests/second
Timeout / response cap	30 sec / 10,485,760 bytes

Why this design

This adapter is live because it has a bounded implementation and governance contract.

Source and reference links

[GLEIF API](https://www.gleif.org/en/lei-data/gleif-api)

<https://www.gleif.org/en/lei-data/gleif-api>

[GLEIF LEI data access](https://www.gleif.org/en/lei-data/access-and-use-lei-data)

<https://www.gleif.org/en/lei-data/access-and-use-lei-data>

SECTION 22

Data source: GDELT DOC 2.0

IMPLEMENTED

Discover topical news metadata and canonical article URLs. The adapter deliberately stores discovery records rather than treating article text as primary evidence.

Catalog ID `gdelt` selects the code-owned connector `gdelt`. The catalog cannot dynamically import arbitrary code. Base URL <https://api.gdeltproject.org> is HTTPS and the adapter is limited to 0.5 requests per second, a 15-second timeout, and 10,485,760 response bytes.

Authentication handling: No credential. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Discovery metadata only. Original publisher terms govern article content; do not treat discovery results as primary evidence.

Query combines legal-name aliases, optional ticker, and configured AI-factory topic terms over a bounded 1-365 day window. Only HTTPS article URLs become discovery documents; metadata includes domain, language, country, tone, and `discovery_only=true`. GDELT result metadata is Tier 3 and cannot by itself support a material financial claim.

Property	Value
Category	news_discovery
Evidence tier	3
Status	implemented
Enabled	True
Rate limit	0.5 requests/second
Timeout / response cap	15 sec / 10,485,760 bytes

Why this design

This adapter is live because it has a bounded implementation and governance contract.

Source and reference links

[GDELT DOC 2.0 API](https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/)

<https://blog.gdeltproject.org/gdelt-doc-2-0-api-debuts/>

SECTION 23

Data source: USAspending.gov

PLANNED

Discover government awards and recipients, then resolve each recipient to the correct public parent before any research use.

Catalog ID usaspending selects the code-owned connector usaspending. The catalog cannot dynamically import arbitrary code. Base URL <https://api.usaspending.gov> is HTTPS and the adapter is limited to 2.0 requests per second, a 45-second timeout, and 20,971,520 response bytes.

Authentication handling: No credential. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Public U.S. federal award data; recipient entity resolution must be reviewed before scoring.

Parse award, recipient, amount, action date, and agency fields; normalize recipient identifiers and preserve modifications. Deduplicate award actions from award totals and preserve the public-parent resolution decision.

Property	Value
Category	government_awards
Evidence tier	1
Status	planned
Enabled	False
Rate limit	2.0 requests/second
Timeout / response cap	45 sec / 20,971,520 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[USAspending API](https://api.usaspending.gov/docs/endpoints)
<https://api.usaspending.gov/docs/endpoints>

SECTION 24

Data source: OpenFIGI

PLANNED

Normalize securities and share classes with FIGI and related identifiers after legal-entity resolution.

Catalog ID openfigi selects the code-owned connector openfigi. The catalog cannot dynamically import arbitrary code. Base URL <https://api.openfigi.com> is HTTPS and the adapter is limited to 0.35 requests per second, a 30-second timeout, and 10,485,760 response bytes.

Authentication handling: header via environment variable OPENFIGI_API_KEY. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Review OpenFIGI terms and identifier redistribution requirements before production use.

Batch mapping requests by identifier type and market sector, preserve all candidates, and require deterministic share-class selection rules. Record terms and redistribution limits before identifiers are shown in exported research.

Property	Value
Category	security_identity
Evidence tier	2
Status	planned
Enabled	False
Rate limit	0.35 requests/second
Timeout / response cap	30 sec / 10,485,760 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[OpenFIGI API](https://www.openfigi.com/api/documentation)

<https://www.openfigi.com/api/documentation>

SECTION 25

Data source: UK Companies House

PLANNED

Add UK official company identity, profiles, filing histories, and document discovery.

Catalog ID companies_house selects the code-owned connector companies_house. The catalog cannot dynamically import arbitrary code. Base URL <https://api.company-information.service.gov.uk> is HTTPS and the adapter is limited to 1.5 requests per second, a 30-second timeout, and 20,971,520 response bytes.

Authentication handling: basic_username via environment variable COMPANIES_HOUSE_API_KEY. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Crown/public register data; observe Companies House API and document-service terms.

Separate company profile, filing history, and document retrieval; preserve company number as the stable official key. Retain filing dates, category, document metadata, and raw content hashes.

Property	Value
Category	regulatory_filings
Evidence tier	1
Status	planned
Enabled	False
Rate limit	1.5 requests/second
Timeout / response cap	30 sec / 20,971,520 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[Companies House API](https://developer.company-information.service.gov.uk/)

<https://developer.company-information.service.gov.uk/>

[Developer guidelines](https://developer.company-information.service.gov.uk/developer-guidelines/)

<https://developer.company-information.service.gov.uk/developer-guidelines/>

SECTION 26

Data source: filings.xbrl.org ESEF/UKSEF index

PLANNED

Discover ESEF and UKSEF structured filings, with material facts checked against the national official record.

Catalog ID filings_xbrl_org selects the code-owned connector filings_xbrl_org. The catalog cannot dynamically import arbitrary code. Base URL <https://filings.xbrl.org> is HTTPS and the adapter is limited to 1.0 requests per second, a 45-second timeout, and 52,428,800 response bytes.

Authentication handling: No credential. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Free public API with change/withdrawal caveat; validate against national OAM records when material.

Use the index for discovery and taxonomy-aware parsing, but retain the national OAM link used to validate material facts. Handle withdrawn and corrected filings as new point-in-time versions.

Property	Value
Category	regulatory_filings
Evidence tier	1
Status	planned
Enabled	False
Rate limit	1.0 requests/second
Timeout / response cap	45 sec / 52,428,800 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[filings.xbrl.org API](https://filings.xbrl.org/docs/api)

<https://filings.xbrl.org/docs/api>

[ESMA ESEF resources](https://www.esma.europa.eu/issuer-disclosure/electronic-reporting)

<https://www.esma.europa.eu/issuer-disclosure/electronic-reporting>

SECTION 27

Data source: Japan EDINET API v2

PLANNED

Add official Japanese filings and structured disclosure packages.

Catalog ID edinet selects the code-owned connector edinet. The catalog cannot dynamically import arbitrary code. Base URL <https://api.edinet-fsa.go.jp> is HTTPS and the adapter is limited to 1.0 requests per second, a 45-second timeout, and 52,428,800 response bytes.

Authentication handling: query via environment variable EDINET_API_KEY. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Official disclosure API; registration and API key are required.

Persist document list metadata before downloading disclosure ZIP packages; preserve EDINET code, filer, form, and filing date. Use an XBRL-aware parser and Japanese taxonomy tests rather than generic tag matching.

Property	Value
Category	regulatory_filings
Evidence tier	1
Status	planned
Enabled	False
Rate limit	1.0 requests/second
Timeout / response cap	45 sec / 52,428,800 bytes

Why this design
The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[EDINET API registration](https://disclosure2.edinet-fsa.go.jp/week0020.aspx)
<https://disclosure2.edinet-fsa.go.jp/week0020.aspx>

SECTION 28

Data source: U.S. Energy Information Administration API v2

PLANNED

Supply power-market context for generation, capacity, fuel, pricing, and infrastructure demand without pretending that market data proves a company award.

Catalog ID eia selects the code-owned connector eia. The catalog cannot dynamically import arbitrary code. Base URL <https://api.eia.gov> is HTTPS and the adapter is limited to 1.0 requests per second, a 45-second timeout, and 20,971,520 response bytes.

Authentication handling: query via environment variable EIA_API_KEY. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Public government energy data; a free API key is required.

Normalize series ID, geography, frequency, unit, period, and revision metadata. Keep market context separate from company evidence unless a project or contract source supplies the linkage.

Property	Value
Category	power_market
Evidence tier	1
Status	planned
Enabled	False
Rate limit	1.0 requests/second
Timeout / response cap	45 sec / 20,971,520 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[EIA API v2](https://www.eia.gov/opensite/openapi/v2/)

<https://www.eia.gov/opensite/openapi/v2/>

SECTION 29

Data source: USPTO PatentsView PatentSearch

PLANNED

Supply patent and assignee evidence as one input to the standards/IP moat component; raw patent counts do not determine the score.

Catalog ID patentsview selects the code-owned connector patentsview. The catalog cannot dynamically import arbitrary code. Base URL <https://search.patentsview.org> is HTTPS and the adapter is limited to 1.0 requests per second, a 45-second timeout, and 20,971,520 response bytes.

Authentication handling: header via environment variable PATENTSVIEW_API_KEY. Credentials are read only at transport time and are omitted from public source metadata, raw manifests, logs, workflow state, and audit payloads. Licence note: Public patent data; PatentSearch API key required. Patent counts alone are not moat evidence.

Resolve assignees to the security master with confidence and ambiguity review. Classify relevant patent families and citations, avoiding duplicate grants and portfolio-size bias.

Property	Value
Category	patents
Evidence tier	2
Status	planned
Enabled	False
Rate limit	1.0 requests/second
Timeout / response cap	45 sec / 20,971,520 bytes

Why this design

The source is catalogued but fails closed until its adapter, fixtures, entity-resolution rules, licensing review, and benchmark tests are complete.

Source and reference links

[PatentsView Search API](https://search.patentsview.org/docs/docs/Search%20API/SearchAPIReference/)

<https://search.patentsview.org/docs/docs/Search%20API/SearchAPIReference/>

SECTION 30

Adding a source without hardwiring research data

IMPLEMENTED

Configuration declares source policy; an allow-listed adapter implements bounded behavior; evidence remains source-derived.

A new source begins disabled. The catalog entry defines the official base URL, endpoints, source tier, authentication environment reference, operational limits, licence notes, and source-specific options. The connector contract defines point-in-time semantics and output type before implementation begins.

The adapter archives the original response before parsing, emits normalized documents or deterministic claims, and records a cursor and sync result. Provider fixtures cover pagination, corrections, malformed responses, rate limits, oversized responses, retries, and credential redaction. Entity and duplicate-document tests follow. Legal, security, data-owner, and model-processing approval precede enablement.

Implementation details

- Create disabled catalog entry.
- Define output and cutoff semantics.
- Implement registry-owned adapter.
- Archive raw bytes and manifest.
- Add parser/entity/idempotency/security tests.
- Run frozen benchmark and rank diff.
- Approve rights and enable explicitly.

Why this design

A URL in configuration is not executable authority. Separating source definition from adapter code prevents a compromised catalog from importing arbitrary code or weakening the evidence contract.

SECTION 31

Bounded HTTPS acquisition

IMPLEMENTED

The source client enforces HTTPS, public DNS destinations, safe redirects, rate limits, retries, response caps, and sanitized metadata.

For each request the client constructs an endpoint from an HTTPS base URL, injects the configured credential immediately before transport, and sets a declared User-Agent. Conditional ETag and Last-Modified headers are applied from the stored cursor. DNS is resolved and every destination is rejected if private, loopback, link-local, reserved, localhost, or .local.

Redirects are disabled in HTTPX and followed manually for at most six hops so every location is revalidated. HTTP 429 and 5xx responses use bounded backoff; total attempts are clamped to one through eight. Only safe response headers are retained. The body is checked against the per-source byte cap before it can enter storage.

Control	Implementation
Scheme	HTTPS only
DNS	Reject non-public addresses
Redirects	Revalidate each hop; maximum six
Retries	Provider option, clamped to 1-8
Rate	Per-source monotonic interval
Headers	Allow list only
Body	Per-source byte ceiling

Concrete contract

```
catalog -> validate URL/DNS -> rate wait -> request -> safe redirect loop
-> retry policy -> response-size check -> sanitized FetchResponse
```

SECTION 32

Content-addressed raw storage

IMPLEMENTED

Every fetched body is written to a deterministic SHA-256 path before downstream interpretation.

RawSnapshotStore hashes the bytes, creates a directory by source/date/hash prefix, and writes the body with a content-type-derived extension. Writes use a temporary file followed by atomic replace. Repeated storage of the same bytes is idempotent because the content path already exists.

A sidecar manifest records schema version, source ID, external ID, source URL, content type, hash, byte count, retrieval timestamp, safe response headers, and adapter metadata. Normalized text is stored beside the original object with a .normalized.txt suffix. The interface is intentionally small so S3 or MinIO can replace local files later.

Concrete contract

```
artifacts/raw-sources/{source_id}/YYYY/MM/DD/{hash[0:2]}/  
{sha256}.{json|html|pdf|txt|bin}  
{sha256}.manifest.json  
{sha256}.normalized.txt # when a text normalizer runs
```

SECTION 33

SEC universe ingestion

IMPLEMENTED

The issuer universe comes from the regulator rather than a code-owned ticker list.

The connector requests `company_tickers_exchange.json` with conditional headers. It fails if the provider schema is not exactly CIK, name, ticker, exchange. Rows are grouped by zero-padded CIK so multiple tickers and exchanges remain attached to one issuer. A stable company ID `sec-{CIK}` and identifier records are upserted.

New universe companies begin ineligible and without a capital-stack segment; a later company sync and research-classification process must complete those fields. Existing research classifications are preserved during identity refresh. The raw universe response and a `source_document` record remain available for audit.

Concrete contract

```
source-sync sec_edgar --mode universe --limit 1000
```

Source and reference links

SEC EDGAR APIs

<https://www.sec.gov/search-filings/edgar-application-programming-interfaces>

SEC fair-access resources

<https://www.sec.gov/about/developer-resources>

SECTION 34

SEC company and filing ingestion

IMPLEMENTED

A company sync joins issuer identity, submissions history, XBRL facts, and bounded recent filing text.

The caller supplies a CIK directly or selects a company whose metadata contains one. The adapter archives submissions and Company Facts, upserts identity and selected security, and creates two source documents. It then chooses allowed forms from the catalog and downloads at most the configured filing limit with filing dates on or before the cutoff.

Each filing response is archived, converted from HTML to normalized text, scanned for prompt-injection signatures, and stored as a `regulatory_filing` document with accession/form metadata. The cursor records the latest accession and last cutoff so operational history and incremental behavior remain visible.

Concrete contract

```
source-sync sec_edgar --mode company --cik 0000320193
--segment compute_servers --subsegment ai_servers
--as-of-date 2026-09-01 --limit 3
```

SECTION 35

Deterministic SEC XBRL parsing

IMPLEMENTED

Revenue and operating margin are calculated from regulator facts without an LLM.

Revenue candidates are searched across RevenueFromContractWithCustomerExcludingAssessedTax, Revenues, and SalesRevenueNet. Only USD facts from 10-K, 20-F, or 40-F filings with fiscal-period FY or null and filed on/before the cutoff are considered. The latest candidate is chosen by period end, filing date, then accession.

OperatingIncomeLoss must match the selected revenue period. Revenue is stored in USD millions at 0.99 confidence. Operating margin is OperatingIncomeLoss divided by revenue times 100 and stored at 0.98 confidence. Both claims point to the XBRL concept or deterministic calculation path and remain Tier 1.

Concrete contract

```
operating_margin_pct = OperatingIncomeLoss / selected_annual_revenue * 100
selection key = (period_end, filed_date, accession)
cutoff rule = filed_date <= research_as_of_date
```

Why this design

Reported financial arithmetic is testable and repeatable. A model is useful for interpreting narrative disclosures, but it should not replace exact tag and period rules where structured regulator facts exist.

SECTION 36

HTML normalization and parsing

IMPLEMENTED

Filing HTML is reduced to stable text while preserving the immutable original response.

BeautifulSoup with the lxml parser reads the archived response. Script, style, noscript, and template nodes are removed; visible text is joined with spaces and whitespace is collapsed. The normalized path is recorded in document metadata, while the content hash continues to identify the original bytes.

This parser is intentionally conservative. It does not claim page coordinates, table-cell lineage, inline XBRL semantics, or rendered-layout fidelity. Those concerns require specialized adapters and fixtures. Model extraction may use the normalized text only after the document passes company, date, size, and local-path checks.

SECTION 37

Normalized evidence-package ingestion

IMPLEMENTED

External parsers can integrate without bypassing the platform by emitting one normalized package contract.

EvidencePackageIngestor requires companies, documents, and claims arrays. Companies are upserted first, documents second, and claims last to satisfy identity and lineage dependencies. Document previews are scanned for prompt-injection patterns and flagged; every claim is converted to the typed EvidenceClaim model before persistence.

The package contract keeps parsing separate from research judgment. A PDF, spreadsheet, API, licensed feed, or future document-AI pipeline may have different extraction mechanics, but it must still produce stable identity, document provenance, atomic claims, units, periods, exact evidence spans, tiers, and cutoff semantics.

Concrete contract

```
{
  companies: [...],
  documents: [{id, company_id, source_url, published_at, content_hash, ...}],
  claims: [{claim_type, value_numeric, unit, period_end, evidence_span, ...}]
}
```

SECTION 38

Model-assisted evidence retrieval

IMPLEMENTED

The extraction service retrieves a bounded, keyword-relevant subset of one company's eligible documents before calling the configured model.

Documents must belong to the company, exist locally, be published on/before the cutoff, and be among the selected document limit. Files larger than 75 MB are rejected. PDFs are explicitly rejected until the optional page-aware parser is implemented. Text, JSON, normalized text, and controlled HTML are supported.

Text is split into 6,000-character chunks with 500-character overlap. Each chunk receives a deterministic keyword score against the extraction-policy claim definitions. The top eight relevant chunks and only their matched claim definitions enter the prompt. The model receives a maximum of 2,400 completion tokens and may return at most eight proposals per document.

Budget	Value
Chunk size	6,000 characters
Overlap	500 characters
Context chunks	Top 8
Proposals/document	Maximum 8
Model completion	Maximum 2,400 tokens
Document size	75 MB

Why this design

Role-specific retrieval reduces latency, cost, prompt-injection surface, and cross-company leakage. It also produces deterministic chunk identifiers that can be checked after generation.

SECTION 39

Proposal validation and human acceptance

IMPLEMENTED

A model response is stored as pending or invalid; it never becomes an evidence claim automatically.

Every proposal must use a claim type allowed for the retrieved context. Its quote must be an exact contiguous 20-600 character substring of the cited chunk. Confidence must be finite and no greater than 0.90. Numeric claims must be finite, within the configured range, and use the exact unit. Narrative claims may not contain numeric values and require nonempty text. Period end must be an ISO date or null.

The proposal ID is a deterministic UUID derived from document ID and normalized proposal JSON. An analyst may accept or reject only a pending proposal. Acceptance creates a second deterministic claim ID, persists the accepted EvidenceClaim with the model and prompt provenance in `extraction_method`, updates proposal review state, and writes an audit event.

Concrete contract

```
MODEL OUTPUT -> deterministic validation -> pending | invalid  
PENDING + analyst accept -> accepted EvidenceClaim + audit event  
PENDING + reject -> retained rejected proposal + audit event
```

Why this design

Human review occurs at the evidence boundary, where the reviewer can still compare the exact source quote, unit, period, and rubric. Waiting until after ranking would hide which model assumption moved the score.

SECTION 40

PDF and rendered-document roadmap

PLANNED

PDF parsing is a deliberate fail-closed boundary in the current implementation, not an unmentioned capability.

The next adapter should store original PDF bytes, extract page-aware text with pypdf or pdfplumber, render pages for visual quote verification, and retain page number, bounding box, table coordinates, OCR status, and parser version. Scanned filings require OCRmyPDF/Tesseract plus confidence and human review.

Tables must preserve row/column lineage and unit headers. Hidden text and prompt-injection content must be compared with the rendered page. The benchmark should include multi-column layouts, footnotes, rotated tables, scanned pages, conflicting unit multipliers, and restatements. Until that adapter exists, the service raises a clear error rather than silently returning unreliable text.

SECTION 41

System-of-record design

IMPLEMENTED

SQLite/WAL is the authoritative local ledger; raw bytes and reports remain content-addressed filesystem artifacts.

Database initialization enables write-ahead logging and foreign keys. Repository methods open short-lived managed connections and use transactions for grouped writes. Unique indexes prevent duplicate security records, duplicate document URL/hash pairs, duplicate run/company assessments, and duplicate run/company rankings.

JSON columns hold flexible metadata and complete structured assessments while first-class columns preserve common filters and integrity. The design favors auditability and low operational friction on one node. PostgreSQL is required before multiple concurrent writers or replicas.

Storage class	Current path	Production target
Transactional ledger	artifacts/aifactory.db	PostgreSQL + migrations
Raw source bytes	artifacts/raw-sources/	Versioned S3/MinIO
Normalized text	Beside raw hash	Versioned object key
Reports/manifests	artifacts/reports/	Immutable object + metadata row
Policy/prompt code	config/	Versioned release artifact

SECTION 42

Database table: companies

IMPLEMENTED

Canonical issuer and selected-security universe, including eligibility, capital-stack placement, public-parent linkage, and source-derived metadata.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
legal_name	TEXT	NOT NULL
ticker	TEXT	NOT NULL
exchange	TEXT	NOT NULL
security_id	TEXT	NOT NULL
segment	TEXT	NOT NULL
subsegment	TEXT	NOT NULL
public_parent_id	TEXT	optional
eligible	INTEGER	NOT NULL, default 1
demo	INTEGER	NOT NULL, default 0
metadata_json	TEXT	NOT NULL, default '{}'
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 43

Database table: market_segments

IMPLEMENTED

Versioned AI-factory taxonomy with reference spend, seed weight, assumptions, and a validation flag.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
name	TEXT	NOT NULL
taxonomy_version	TEXT	NOT NULL
reference_spend_usd_billions	REAL	NOT NULL
reference_weight	REAL	NOT NULL
validated	INTEGER	NOT NULL, default 0
assumptions_json	TEXT	NOT NULL, default '{}'
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 44

Database table: source_documents

IMPLEMENTED

Immutable provenance catalog for every archived or normalized source document.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
company_id	TEXT	optional
source_type	TEXT	NOT NULL
source_tier	TEXT	NOT NULL
publisher	TEXT	NOT NULL
title	TEXT	NOT NULL
source_url	TEXT	NOT NULL
published_at	TEXT	NOT NULL
retrieved_at	TEXT	NOT NULL
content_hash	TEXT	NOT NULL
local_path	TEXT	optional
licence_notes	TEXT	optional
parser_version	TEXT	optional
injection_flags_json	TEXT	NOT NULL, default '[]'
metadata_json	TEXT	NOT NULL, default '{}'

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 45

Database table: evidence_claims

IMPLEMENTED

Atomic accepted facts, calculations, assumptions, and narratives used by the research graph.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
company_id	TEXT	NOT NULL
document_id	TEXT	NOT NULL
claim_type	TEXT	NOT NULL
value_numeric	REAL	optional
value_text	TEXT	optional
unit	TEXT	optional
period_end	TEXT	optional
confidence	REAL	NOT NULL
evidence_span	TEXT	NOT NULL
page_or_section	TEXT	NOT NULL
source_tier	TEXT	NOT NULL
published_at	TEXT	NOT NULL
as_of_eligible	INTEGER	NOT NULL, default 1
contradiction	INTEGER	NOT NULL, default 0
extraction_method	TEXT	NOT NULL, default 'structured'
created_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 46

Database table: entity_identifiers

IMPLEMENTED

CIK, ticker, LEI, and future identifier mappings with source and confidence.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
company_id	TEXT	PK, NOT NULL
scheme	TEXT	PK, NOT NULL
value	TEXT	PK, NOT NULL
source_id	TEXT	NOT NULL
confidence	REAL	NOT NULL, default 1.0
metadata_json	TEXT	NOT NULL, default '{}'
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 47

Database table: source_cursors

IMPLEMENTED

Incremental and conditional-fetch state per source and logical scope.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
source_id	TEXT	PK, NOT NULL
scope	TEXT	PK, NOT NULL
cursor_json	TEXT	NOT NULL, default '{}'
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 48

Database table: source_sync_runs

IMPLEMENTED

Operational history of every connector attempt, including before/after cursors, counters, status, and sanitized error.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
source_id	TEXT	NOT NULL
scope	TEXT	NOT NULL
status	TEXT	NOT NULL
cursor_before_json	TEXT	NOT NULL, default '{}'
cursor_after_json	TEXT	NOT NULL, default '{}'
counters_json	TEXT	NOT NULL, default '{}'
error	TEXT	optional
started_at	TEXT	NOT NULL
completed_at	TEXT	optional

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 49

Database table: claim_proposals

IMPLEMENTED

Model-proposed evidence awaiting explicit review, including validation failures and model/prompt versions.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
company_id	TEXT	NOT NULL
document_id	TEXT	NOT NULL
claim_type	TEXT	NOT NULL
payload_json	TEXT	NOT NULL
status	TEXT	NOT NULL
validation_errors_json	TEXT	NOT NULL, default '[]'
model_provider	TEXT	NOT NULL
model_name	TEXT	NOT NULL
prompt_version	TEXT	NOT NULL
reviewer	TEXT	optional
review_comment	TEXT	optional
reviewed_at	TEXT	optional
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 50

Database table: research_runs

IMPLEMENTED

Frozen research clock and configuration manifest for a portfolio execution.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
as_of_date	TEXT	NOT NULL
status	TEXT	NOT NULL
taxonomy_version	TEXT	NOT NULL
scoring_version	TEXT	NOT NULL
prompt_version	TEXT	NOT NULL
model_provider	TEXT	NOT NULL
model_name	TEXT	NOT NULL
config_snapshot_json	TEXT	NOT NULL
started_at	TEXT	optional
completed_at	TEXT	optional
published_at	TEXT	optional
error	TEXT	optional
created_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 51

Database table: company_assessments

IMPLEMENTED

Complete structured per-company dossier and the rankability/review state used at portfolio fan-in.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
run_id	TEXT	NOT NULL
company_id	TEXT	NOT NULL
rankable	INTEGER	NOT NULL
review_required	INTEGER	NOT NULL
review_status	TEXT	NOT NULL
evidence_confidence	REAL	NOT NULL
citation_coverage	REAL	NOT NULL
base_tafgs	REAL	NOT NULL
risk_adjusted_tafgs	REAL	NOT NULL
assessment_json	TEXT	NOT NULL
created_at	TEXT	NOT NULL
updated_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 52

Database table: rankings

IMPLEMENTED

Base rank plus bear and bull ranks, risk-adjusted score, and scenario-aware rank confidence.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
run_id	TEXT	PK, NOT NULL
company_id	TEXT	PK, NOT NULL
rank	INTEGER	NOT NULL
bear_rank	INTEGER	NOT NULL
bull_rank	INTEGER	NOT NULL
risk_adjusted_tafgs	REAL	NOT NULL
rank_confidence	REAL	NOT NULL
created_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 53

Database table: reviews

IMPLEMENTED

Human dossier decisions, comments, and recorded override metadata.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
run_id	TEXT	NOT NULL
company_id	TEXT	NOT NULL
reviewer	TEXT	NOT NULL
decision	TEXT	NOT NULL
comment	TEXT	NOT NULL
overrides_json	TEXT	NOT NULL, default '{}'
created_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 54

Database table: audit_events

IMPLEMENTED

Append-only governance and operational events scoped to a run and/or company.

This table is part of the implemented 13-table SQLite schema. Its columns below are read from the actual artifacts/aifactory.db schema used by the platform, not a conceptual future model.

Rows are retained as operational or governance history. Updates are idempotent where identity is stable, while research runs, reviews, sync attempts, and audit events preserve chronological state rather than hiding earlier outcomes.

Column	Type	Constraint / default
id	TEXT	PK
run_id	TEXT	optional
company_id	TEXT	optional
actor	TEXT	NOT NULL
event_type	TEXT	NOT NULL
payload_json	TEXT	NOT NULL
created_at	TEXT	NOT NULL

Why this design

The table has one bounded ownership purpose so identity, provenance, accepted evidence, model proposals, assessments, rankings, reviews, and audit history do not collapse into an ungoverned document store.

SECTION 55

Artifact paths and provenance example

IMPLEMENTED

A reviewer can travel from a ranked field back to an accepted claim, source document, raw bytes, and retrieval manifest.

The database stores `evidence_claim.document_id`. The document record stores `source_url`, `published_at`, `retrieved_at`, `content_hash`, `local_path`, `licence note`, `parser version`, `injection flags`, and `metadata`. The `local_path` resolves to the SHA-256 raw object; the adjacent manifest repeats the retrieval facts and safe response headers.

A report manifest stores run versions, model route, ranked-company count, report hash, and the union of evidence claim IDs. This creates a reproducibility chain from publication to score, score to assessment, assessment to claim, claim to document, and document to immutable bytes.

Concrete contract

```
report.manifest.json
-> research_runs + rankings + company_assessments
-> evidence_claim_ids
-> evidence_claims.document_id
-> source_documents.local_path + content_hash
-> raw object + sidecar manifest + source_url
```

SECTION 56

Data lifecycle and retention

IMPLEMENTED

Source acquisition, proposal review, research runs, and publication create separate lifecycle records.

A connector attempt begins in `source_sync_runs`, captures `cursor_before`, and ends `completed`, `not_modified`, `failed`, or `interrupted` with counters and `cursor_after`. Raw objects remain immutable. A parsed document can yield deterministic claims or model proposals. Invalid and rejected proposals remain visible; accepted proposals create claims without deleting their proposal history.

A research run freezes versions and company IDs, creates per-company assessments, and writes rankings after fan-in. Dossier reviews are separate rows. Publication changes the run status and regenerates report artifacts only after the approval gate. Corrections create a new source version and new run, preserving the old record.

Concrete contract

```
FETCH HISTORY -> RAW VERSION -> DOCUMENT -> PROPOSAL -> REVIEW -> CLAIM  
RUN CREATED -> RUNNING -> COMPLETED -> REVIEWED -> PUBLISHED  
CORRECTION -> NEW DOCUMENT HASH + NEW RUN (never history rewrite)
```

SECTION 57

Volume I implementation checklist

IMPLEMENTED

Use this checklist to verify that data entered the platform through the intended control path.

Implementation details

- The source exists in config/sources.json and names a code-owned adapter.
- HTTPS destination, DNS, redirect, rate, retry, timeout, and byte limits passed.
- The original response and sidecar manifest exist under the content hash.
- The source_document contains company, source URL, dates, hash, parser, licence, and flags.
- Identity resolution preserves stable identifiers and ambiguity evidence.
- Reported financials came from Tier-1 deterministic parsing.
- Model output is a pending/invalid proposal until explicit review.
- Accepted claims include type, value/text, unit, period, exact span, section, confidence, tier, and date.
- The run cutoff excludes later evidence and the report manifest resolves every claim ID.

